

LOCAL LERANING COEFFICIENTS AND GROUP INVARIANT NETWORKS NOTES

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As shown in [13], deep neural networks are singular models, i.e. the loss landscape of a deep neural network may contain local minima at which the Hessian is degenerate. Consequently, several classical results for regular models, such as asymptotic normality and the Bayesian Information Criterion, are no longer directly applicable. This motivates the use of Singular Learning Theory to study model selection, generalization and related questions in neural networks, as well as developed quantities, such as patterning or susceptibility, in AI alignment (see [3, 5, 11]).

One important source of such singular behavior is the presence of symmetries in the parameter space. Indeed, many neural network architectures possess hidden group symmetries so that the loss values of critical points on the same group orbit are identical. [1] proves that ReLU MLPs of arbitrary depth and non-increasing widths admit only scaling and neuron-permutation symmetries. In a two-layer ReLU model, [2] classifies all symmetries of the true parameters and shows that they consist only of positive scaling $\mathbb{R}_{>0}$, neuron permutations S_k , and local orientation reversal $\mathbb{Z}_2$, with the first two being generic. Related symmetry structures may also arise in architectures such as transformers. More broadly, neural networks can possess richer and more complex group symmetries, particularly in equivariant neural network where the group action is imposed *a priori*, including continuous group $O(3)$, etc..

In this note, we mainly aim to explore two aspects: symmetry and singularity, as well as their interaction. More precisely, we consider the following questions:

- (1) Under what assumptions is the output of a neural network G -invariant, the associated loss function G -invariant, and thereby the gradient of the loss G -equivariant? This provides the basic link with tools like Equivariant Degree Theory.
- (2) How do such symmetries influence singular-learning quantities, such as local learning coefficient, free energy, etc.?

In section 1, we set up the statistical framework. Section 2 discusses MLPs with permutation symmetries and formulates conditions under which such G -invariance occur. Section 3.1 recalls key definitions of Singular Learning Theory (KL divergence, volume function, learning coefficient and its local version, estimated LLC, free energy and local free energy, as well as its asymptotic expansion. Section 3.2 shows that, under certain conditions, the local learning coefficient (LLC) of a G -invariant loss and hence the estimated LLC and the local free energy are G -invariant. An example of MLPs with permutation symmetries is presented in 4 to further illustrate these G -invariance results. In the Appendix A, we recalls Watanabe's Theorem 7.1 and the part of its proof relevant to the present discussion (One is referred to [12] for more details of this theorem).

1. STATISTICAL SETTING

Let $D_n = \{(x_i, y_i)\}_{i=1}^n$ be i.i.d. sampled from a true distribution $q(x, y) = q(y | x) q(x)$, where $x_i \in \mathbb{R}^s$, $y_i \in \mathbb{R}^m$, and the marginal $q(x)$ is assumed known. We model $q(y | x)$ by a parametric

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$$p(y \mid x, \mathbf{w}), \quad \mathbf{w} \in W.$$

Here, W denotes the parameter space and the specific form of $\mathbf{w}$ in the MLP setting will be given in Section 2. We assume that W is equipped with a prior density $\varphi(w)$ satisfying $\varphi(w) > 0$.

2. MLPs AND THEIR HIDDEN SYMMETRIES.

In this section we discuss the conditions under which neuron permutations leave a fully connected MLP, and its associated loss, invariant. These conditions may involve the input distribution, the teacher model, the activation function, as well as an examination of whether the same arguments extend to other settings, such as teacher-student model.

Assume a fully connected network with L hidden layers. Let $k_0 = s$ be the input dimension, $k_1, \dots, k_L$ be the hidden-layer widths, and $k_{L+1} = m$ be the output dimension. There we have the weight tuple

$$\mathbf{w} = (\mathbf{w}^{(1)}, \mathbf{w}^{(2)}, \dots, \mathbf{w}^{(L)}, \mathbf{w}^{(L+1)}),$$

where $\mathbf{w}^{(l)} \in \mathbb{R}^{k_l \times k_{l-1}}$, $l = 1, \dots, L+1$. (Bias terms are omitted here but will be included shortly)

Define the population loss

$$(1) \quad L(\mathbf{w}) := \mathbb{E}_{(x,y) \sim q} \|f(x, \mathbf{w}) - y\|^2,$$

where $f(x, \mathbf{w})$ denotes the model output. The corresponding empirical loss is obtained by averaging over the sample D_n .

Theorem 1. *Let*

$$G = S_{k_1} \times S_{k_2} \times \dots \times S_{k_L}.$$

For $g = (P_1, \dots, P_L) \in G$ where P_l is the permutation matrix corresponding to S_{k_l} . We define the G -action on the tuple $\mathbf{w}$ by

$$(2) \quad (g \cdot \mathbf{w})^{(l)} := P_l \mathbf{w}^{(l)} P_{l-1}^{-1}, \quad l = 1, \dots, L+1$$

where $P_0 = I_s$ and $P_{L+1} = I_m$. If activation σ is pointwise, then both the network f and its loss $L(\mathbf{w})$ are G -invariant.

Proof. Let $h_0(x) = x$ and $h_l(x) = \sigma(\mathbf{w}^{(l)} h_{l-1}(x))$, $l = 1, \dots, L$. Suppose the network output is

$$f(x, \mathbf{w}) = \mathbf{w}^{(L+1)} h_L(x).$$

For $g = (P_1, \dots, P_L) \in G$, let $\tilde{\mathbf{w}} = g\mathbf{w}$, $\tilde{h}_0(x) = x$ and $\tilde{h}_l(x) = \sigma(\tilde{\mathbf{w}}^{(l)} \tilde{h}_{l-1}(x))$, one has

$$\tilde{h}_l(x) = P_l h_l(x).$$

Indeed, for the base case $l = 1$,

$$\tilde{h}_1(x) = \sigma(\tilde{\mathbf{w}}^{(1)} x) = \sigma(P_1 \mathbf{w}^{(1)} x) = P_1 \sigma(\mathbf{w}^{(1)} x) = P_1 h_1(x).$$

Now assume $\tilde{h}_{l-1}(x) = P_{l-1} h_{l-1}(x)$, then

$$\begin{aligned} \tilde{h}_l(x) &= \sigma(\tilde{\mathbf{w}}^{(l)} \tilde{h}_{l-1}(x)) = \sigma(P_l \mathbf{w}^{(l)} P_{l-1}^{-1} P_{l-1} h_{l-1}(x)) \\ &= \sigma(P_l \mathbf{w}^{(l)} h_{l-1}(x)) = P_l \sigma(\mathbf{w}^{(l)} h_{l-1}(x)) \\ &= P_l h_l(x), \quad l = 1, \dots, L. \end{aligned}$$

So for the output layer

$$\tilde{\mathbf{w}}^{(L+1)} = P_{L+1} \mathbf{w}^{(L+1)} P_L^{-1} = \mathbf{w}^{(L+1)} P_L^{-1},$$

we have

$$\begin{aligned} f(x, g \cdot \mathbf{w}) &= \tilde{\mathbf{w}}^{(L+1)} \tilde{h}_L(x) = \mathbf{w}^{(L+1)} P_L^{-1} P_L h_L(x) \\ &= \mathbf{w}^{(L+1)} h_L(x) \\ &= f(x, \mathbf{w}). \end{aligned}$$

Accordingly, $L(g \cdot \mathbf{w}) = L(\mathbf{w})$ □

Remark 2. We emphasize that

- (1) The group G acts by row and column permutations at each hidden layer.
- (2) The invariance condition is quite flexible. The only requirement is activation function σ being pointwise. More precisely,
 - (a) No symmetry assumption on the input distribution $x \sim q(x)$ is needed.
 - (b) This G -invariance property also applies to teacher-student setting where teacher model can be arbitrary.

Remark 3. If we further extend our framework to allow permutations of the input layer (as setting in [8, 7]), i.e.

$$G' = S_{k_0} \times S_{k_1} \times \cdots \times S_{k_L},$$

then the network satisfies:

$$f(x, g \cdot \mathbf{w}) = f(P_0^{-1} x, \mathbf{w}), \quad P_0 \in S_{k_0}.$$

Notice that this G' -invariance cannot be applied to loss of supervised learning settings but can be applied to teacher student model under more restrictive conditions:

- (1) Activation σ is pointwise.
- (2) S_{k_0} -Invariance assumption is needed on the input distribution $x \sim q(x)$. One example is $x \sim \mathcal{N}(0, \mathbb{I}_n)$, or $x_i, i = 1, \dots, k_0$ is i.i.d., to mention a few.
- (3) The teacher model must satisfy S_{k_0} input invariance assumptions.

The proof follows a very similar argument to Theorem 1 and will not be repeated here (see Appendix A if needed).

3. THEORETICAL LEARNING COEFFICIENT AND GROUP INVARIANCE

We recall several key concepts from Singular Learning Theory, including the learning coefficient, the local learning coefficient (LLC), the estimated LLC, the free energy, and local free energy. We then prove, under suitable conditions, the G -invariance of several SLT quantities. For further details on the LLC and its variants, see [4, 9, 10].

3.1. Key concepts.

Definition 4 (Learning Coefficient). The volume function is defined by

$$(3) \quad \text{vol}(\epsilon) := \int_{K(\mathbf{w}) < \epsilon} \varphi(\mathbf{w}) d\mathbf{w},$$

where

$$K(\mathbf{w}) = \int \int_{\mathbb{R}^{s+m}} q(y|x) q(x) \log \frac{q(y|x)}{q(y|x, \mathbf{w})} dx dy$$

is called Kullback-Leibler Divergence. Then the learning coefficient is

$$(4) \quad \lambda = - \lim_{\epsilon \rightarrow 0} \log_2 \left[\frac{\text{vol}(\frac{\epsilon}{2})}{\text{vol}(\epsilon)} \right].$$

We next introduce the local version of this quantity.

Definition 5 (Local Learning Coefficient). *Let $\mathbf{w}^*$ be local minima of $K(\mathbf{w})$, and let $B(\mathbf{w}^*)$ be the neighborhood of $\mathbf{w}^*$ where $K(\mathbf{w}) \geq K(\mathbf{w}^*)$. Define*

$$(5) \quad \text{vol}(\epsilon, \mathbf{w}^*) := \int_{\substack{\mathbf{w} \in B(\mathbf{w}^*) \\ |K(\mathbf{w}) - K(\mathbf{w}^*)| < \epsilon}} \varphi(\mathbf{w}) d\mathbf{w}.$$

The local learning coefficient at $\mathbf{w}^$ is*

$$(6) \quad \lambda(\mathbf{w}^*) = -\lim_{\epsilon \rightarrow 0} \log_2 \left[\frac{\text{vol}(\frac{\epsilon}{2}, \mathbf{w}^*)}{\text{vol}(\epsilon, \mathbf{w}^*)} \right].$$

Definition 6 (Estimated Local Learning Coefficient). *We define the estimated local learning coefficient at $\mathbf{w}^*$ to be*

$$(7) \quad \hat{\lambda}(\mathbf{w}^*) = n\beta \left(\mathbb{E}_{\mathbf{w}|\mathbf{w}^*, \gamma}^\beta [L_n(\mathbf{w})] - L_n(\mathbf{w}^*) \right),$$

where $\mathbb{E}_{\mathbf{w}|\mathbf{w}^, \gamma}^\beta$ denotes expectation with respect to this Gibbs posterior*

$$(8) \quad p(\mathbf{w}; \mathbf{w}^*, \beta, \gamma) \propto \exp \left\{ -n\beta L_n(\mathbf{w}) - \frac{\gamma}{2} \|\mathbf{w} - \mathbf{w}^*\|_2^2 \right\}.$$

Here, β is the inverse temperature controlling the contribution of the loss, and γ is the localization strength controlling proximity to $\mathbf{w}^$. (Do we have to have L_n here to be negative-log likelihood loss?)*

Definition 7 (Free Energy). *The free energy F_n is defined as the negative log of the marginal likelihood:*

$$(9) \quad F_n := -\log \int_W \exp(-nL_n^{\text{nl}}(\mathbf{w})) \varphi(\mathbf{w}) d\mathbf{w},$$

where

$$L_n^{\text{nl}}(\mathbf{w}) := -\frac{1}{n} \sum_{i=1}^n \log p(y_i | x_i, \mathbf{w})$$

is the empirical negative log-likelihood and $D_n = \{(x_i, y_i)\}_{i=1}^n$.

By Watanabe, the asymptotic expansion in n takes the form

$$(10) \quad F_n = nL_n^{\text{nl}}(\mathbf{w}^*) + \lambda \log n + O_p(\log \log n), \quad n \rightarrow \infty$$

where $\mathbf{w}^$ is a global minimum (?).*

Definition 8 (Local Free Energy). *The local free energy at $\mathbf{w}^*$ is defined by*

$$(11) \quad F_n(B_\gamma(\mathbf{w}^*)) := -\log \int_{B_\gamma(\mathbf{w}^*)} \exp(-nL_n^{\text{nl}}(\mathbf{w})) \varphi(\mathbf{w}) d\mathbf{w},$$

where $B_\gamma(\mathbf{w}^)$ denotes the ball of radius γ centered at $\mathbf{w}^*$.*

Given a local minimum $\mathbf{w}^*$ of the population loss $L^{\text{nl}}(\mathbf{w})$, SLT gives the following asymptotic expansion:

$$(12) \quad F_n(B_\gamma(\mathbf{w}^*)) = nL_n^{\text{nl}}(\mathbf{w}^*) + \lambda(\mathbf{w}^*) \log n + O_p(\log \log n), \quad n \rightarrow \infty,$$

where $\lambda(\mathbf{w}^)$ is the theoretical LLC.*

3.2. G -Invariance of LLC, Estimated LLC and Local Free Energy. In the following, we prove the G -invariance of the LLC, local free energy and estimated LLC under suitable conditions.

Theorem 9. *Let G be a group acting smoothly on the parameter space W . Suppose $K(\mathbf{w})$ is G -invariant, then local learning coefficient is also G -invariant, i.e.*

$$\lambda(g \cdot \mathbf{w}^*) = \lambda(\mathbf{w}^*), \quad g \in G.$$

Proof. This is essentially the invariance of the LLC under local diffeomorphisms (see Appendix C of [6]). Since $\varphi(\mathbf{w}^*) > 0$, the prior $\varphi(\mathbf{w})$ does not affect the asymptotic exponent in $\text{vol}(\epsilon, \mathbf{w}^*)$. We therefore omit it in the below. By equation (5) and G -invariance of $K(\mathbf{w})$, one has

$$\begin{aligned} \text{vol}(\epsilon, g \cdot \mathbf{w}^*) &= \int_{\substack{\mathbf{w} \in gB(g\mathbf{w}^*) \\ |K(\mathbf{w}) - K(g\mathbf{w}^*)| < \epsilon}} d\mathbf{w} \\ &\stackrel{\mathbf{w} = g\mathbf{u}}{=} \int_{\substack{\mathbf{u} \in B(\mathbf{w}^*) \\ |K(\mathbf{u}) - K(\mathbf{w}^*)| < \epsilon}} |\det D(g\mathbf{u})| d\mathbf{u} \end{aligned}$$

Since G acts smoothly, $u \mapsto g\mathbf{u}$ is diffeomorphism. Thus, there exist c_1, c_2 s.t. $c_1 \leq |\det D(g\mathbf{u})| \leq c_2$, i.e. the Jacobian factor changes the local volume only by a positive bounded factor, and therefore does not affect the asymptotic exponent. By equation (6), we conclude that

$$\lambda(g \cdot \mathbf{w}^*) = -\lim_{\epsilon \rightarrow 0} \log_2 \left[\frac{\text{vol}(\frac{\epsilon}{2}, g \cdot \mathbf{w}^*)}{\text{vol}(\epsilon, g \cdot \mathbf{w}^*)} \right] = -\lim_{\epsilon \rightarrow 0} \log_2 \left[\frac{\text{vol}(\frac{\epsilon}{2}, \mathbf{w}^*)}{\text{vol}(\epsilon, \mathbf{w}^*)} \right] = \lambda(\mathbf{w}^*).$$

□

Theorem 10. *Assume that empirical loss L_n is G -invariant and action of G preserve Euclidean distance. Then the estimated LLC is G -invariant.*

$$\hat{\lambda}(g \cdot \mathbf{w}^*) = \hat{\lambda}(\mathbf{w}^*)$$

Proof. Since $L_n(g \cdot \mathbf{w}) = L_n(\mathbf{w})$ and $\|g \cdot \mathbf{w} - g \cdot \mathbf{w}^*\|^2 = \|\mathbf{w} - \mathbf{w}^*\|^2$, we have

$$\begin{aligned} p(g \cdot \mathbf{w}; g \cdot \mathbf{w}^*, \beta, \gamma) &\propto \exp \left\{ -n\beta L_n(g \cdot \mathbf{w}) - \frac{\gamma}{2} \|g \cdot \mathbf{w} - g \cdot \mathbf{w}^*\|_2^2 \right\} \\ &= \exp \left\{ -n\beta L_n(\mathbf{w}) - \frac{\gamma}{2} \|\mathbf{w} - \mathbf{w}^*\|_2^2 \right\}. \end{aligned}$$

Hence the corresponding normalized Gibbs posteriors coincide, i.e.

$$\mathbb{E}_{g \cdot \mathbf{w} | g \cdot \mathbf{w}^*, \gamma}^\beta [L_n(\mathbf{w})] = \mathbb{E}_{\mathbf{w} | \mathbf{w}^*, \gamma}^\beta [L_n(\mathbf{w})].$$

Therefore, $\hat{\lambda}(g \cdot \mathbf{w}^*) = \hat{\lambda}(\mathbf{w}^*)$. □

Theorem 11. *Let a group G acts smoothly on the parameter space W . Suppose the model is G -invariant in the sense*

$$p(y|x, \mathbf{w}) = p(y|x, g \cdot \mathbf{w}), \quad g \in G$$

for all (x, y) and $\mathbf{w} \in W$. Then the asymptotic expansion of local free energy is invariant under G action.

Proof. Conclusion follows directly from Equation (12) and Theorem 9. □

4. EXAMPLE

Example 12. We now return to the neural network considered in Theorem 1. Let

$$G = S_{k_1} \times S_{k_2} \times \cdots \times S_{k_n}$$

act on the parameter space W by the action defined in (2). Notice that this action is distance-preserving and smooth. Suppose the supervised model is given by

$$y = f(x, \mathbf{w}) + \delta, \quad \delta \sim \mathcal{N}(0, I_m).$$

Then

$$p(y | x, \mathbf{w}) = \frac{1}{(2\pi)^{m/2}} \exp\left(-\frac{1}{2}\|y - f(x, \mathbf{w})\|^2\right),$$

and the MSE loss agrees with the negative log-likelihood up to an additive constant.

By Theorem 1, the network $f(x, \mathbf{w})$ is G -invariant. Hence,

$$p(y | x, g \cdot \mathbf{w}) = p(y | x, \mathbf{w}), \quad g \in G.$$

It follows that the empirical MSE loss L_n , the empirical negative log-likelihood L_n^{nl} , and the KL divergence K are all G -invariant, i.e.

$$L_n(g \cdot \mathbf{w}) = L_n(\mathbf{w}), \quad L_n^{\text{nl}}(g \cdot \mathbf{w}) = L_n^{\text{nl}}(\mathbf{w}), \quad K(g \cdot \mathbf{w}) = K(\mathbf{w}).$$

Therefore, by theorems 9, 10 and 11, the LLC, the estimated LLC, and the asymptotic expansion of F_n are all G -invariant. That is equivalent to say, parameters $\mathbf{w}$ on the same G orbit have the same accuracy and asymptotic complexity.

This raises a question: *how does the local F_n behave at local minima with different isotropy groups? Or, does the local F_n vary across different isotropy strata?* To explore this curiosity, I might shortly look at a two-layer width-six toy model considered in [7]. In this model, local minima with isotropy groups $S_4 \times S_2$ and $S_3 \wr S_2$ can be identified, for example in the forms

$$\mathbf{w}_{S_4 \times S_2}^* = \begin{pmatrix} a & b & b & b & e & e \\ b & a & b & b & e & e \\ b & b & a & b & e & e \\ b & b & b & a & e & e \\ f & f & f & f & c & d \\ f & f & f & f & d & c \end{pmatrix}, \quad \mathbf{w}_{S_3 \wr S_2}^* = \begin{pmatrix} a & b & b & c & c & c \\ b & a & b & c & c & c \\ b & b & a & c & c & c \\ c & c & c & a & b & b \\ c & c & c & b & a & b \\ c & c & c & b & b & a \end{pmatrix},$$

where $a, b, c, d, e, f \in \mathbb{R}$ are pairwise distinct. It is worth noting that, for fixed-point subspaces of intransitive type $S_{k-p} \times S_p$, $p > 1$, and imprimitive type $S_p \wr S_q$, $k = pq$, the dimensions are always 6 and 3, respectively, for arbitrary neuron width k . We then may use the Gibbs posterior in Definition 6 to estimate the corresponding LLCs.

5. SOME FUTURE DISCUSSION

Another aspect that I am curious is the quantities patterning and susceptibility, and to investigate if they admit any meaningful connections with symmetries in MLPs (or transformers). In addition, the proof of Watanabe's Theorem 7.1 in [12] relies on a substantial amount of preceding machinery that I have not yet explored in full. A more careful study of these prerequisites should help make the "suitable conditions" stated in Section 3.2 more precise. I have also briefly study the equivariance properties of equivariant neural networks and their loss. Though the observations appear rather elementary so far, it would be interesting to see if they may lead to more meaningful insights.

APPENDIX A. SOME PROOFS

We next provide details about volume and learning coefficients. One is referred to Watanabe Theorem 7.1 in [12] for more details.

Theorem 13 (Watanabe09 Theorem 7.1). *Assume that the set of true parameters $\{\mathbf{w} \in \text{supp } \varphi; K(\mathbf{w}) = 0\} \neq \emptyset$. Then*

(1) *There exists a constant $c_2 > 0$ such that*

$$\lim_{t \rightarrow 0} \frac{1}{t^{\lambda-1}(-\log t)^{m-1}} \int \delta(t - K(\mathbf{w})) \varphi(\mathbf{w}) d\mathbf{w} = c_2.$$

(2) *Let $V(t)$ be a volume function,*

$$V(t) = \int_{K(\mathbf{w}) < t} \varphi(\mathbf{w}) d\mathbf{w}.$$

For an arbitrary $a > 0$ ($a \neq 1$),

$$\lambda = \lim_{t \rightarrow 0} \frac{\log\{V(at)/V(t)\}}{\log a}.$$

Proof. Let $v(t)$ be the state density function $v(t) = \int \delta(t - K(\mathbf{w})) \varphi(\mathbf{w}) d\mathbf{w}$. By (1), there exists $c_2 > 0$ such that

$$v(t) = c_2 t^{\lambda-1}(-\log t)^{m-1} + o(t^{\lambda-1}(-\log t)^{m-1}).$$

By definition,

$$V(t) = \int_0^t v(s) ds = c_2 \int_0^t s^{\lambda-1}(-\log s)^{m-1} ds + o(t^{\lambda}(-\log t)^{m-1}).$$

By integration by parts, let $u = (-\log s)^{m-1}$, $dv = s^{\lambda-1} ds$. Then

$$du = -(m-1)(-\log s)^{m-2} \frac{1}{s} ds, \quad v = \frac{s^{\lambda}}{\lambda}.$$

Therefore

$$\begin{aligned} \int_0^t s^{\lambda-1}(-\log s)^{m-1} ds &= \frac{t^{\lambda}}{\lambda}(-\log t)^{m-1} + \frac{m-1}{\lambda} \int_0^t s^{\lambda-1}(-\log s)^{m-2} ds \\ &= \frac{1}{\lambda} t^{\lambda}(-\log t)^{m-1} + o(t^{\lambda}(-\log t)^{m-1}). \end{aligned}$$

Hence $V(t) = \frac{c_2}{\lambda} t^{\lambda}(-\log t)^{m-1} + o(t^{\lambda}(-\log t)^{m-1})$, and

$$\frac{V(at)}{V(t)} = \frac{(at)^{\lambda}}{t^{\lambda}} \left[\frac{(-\log at)^{m-1}}{(-\log t)^{m-1}} \right] = a^{\lambda} \left[1 + \frac{\log a}{\log t} \right]^{m-1}.$$

i.e.

$$\lim_{t \rightarrow 0} \frac{V(at)}{V(t)} = a^{\lambda} \lim_{t \rightarrow 0} \left[1 + \frac{\log a}{\log t} \right]^{m-1} = a^{\lambda}.$$

□

Theorem 14. *Let*

$$G' = S_{k_0} \times S_{k_1} \times S_{k_2} \times \cdots \times S_{k_L}.$$

For $g = (P_0, P_1, \dots, P_L) \in G'$ where P_l is the permutation matrix corresponding to S_{k_l} . We define the action on the tuple $\mathbf{w}$ by

$$(g \cdot \mathbf{w})^{(l)} = P_l \mathbf{w}^{(l)} P_{l-1}^{-1}, \quad l = 1, \dots, L+1$$

where $P_{L+1} = I_m$. Here, we consider the teacher-student model with the loss:

$$(13) \quad L_{ts}(\mathbf{w}) := \mathbb{E}_{x \sim q(x)} \|f(x, \mathbf{w}) - f_T(x, \mathbf{w}_T)\|^2,$$

where f_T is the teacher model. Then $f(x, g \cdot \mathbf{w}) = f(P_0^{-1}x, \mathbf{w})$, $P_0 \in S_{k_0}$ and $L_{ts}(\mathbf{w})$ is G' -invariant under the following assumptions:

- (1) Activation σ is pointwise.
- (2) Input distribution $x \sim q(x)$ is S_{k_0} -invariant.
- (3) The teacher model f_T is invariant under input permutations S_{k_0} , namely

$$f_T(P_0x, \mathbf{w}_T) = f_T(x, \mathbf{w}_T), \text{ for all } P_0 \in S_{k_0}.$$

Proof. Let $h_0(x) = x$ and $h_l(x) = \sigma(\mathbf{w}^{(l)} h_{l-1}(x))$ for $l = 1, \dots, L$. Assume the network output

$$f(x, \mathbf{w}) = \mathbf{w}^{(L+1)} h_L(x).$$

For $g = (P_1, \dots, P_L) \in G'$, denote $\tilde{\mathbf{w}} = g\mathbf{w}$. Define $\tilde{h}_0(x) = x$ and $\tilde{h}_l(x) = \sigma(\tilde{\mathbf{w}}^{(l)} \tilde{h}_{l-1}(x))$. We claim that

$$\tilde{h}_l(x) = P_l h_l(P_0^{-1}x).$$

Indeed, for $l = 1$, we have

$$\tilde{h}_1(x) = \sigma(\tilde{\mathbf{w}}^{(1)} x) = \sigma(P_1 \mathbf{w}^{(1)} P_0^{-1} x) = P_1 \sigma(\mathbf{w}^{(1)} P_0^{-1} x) = P_1 h_1(P_0^{-1} x).$$

Assume that $\tilde{h}_{l-1}(x) = P_{l-1} h_{l-1}(P_0^{-1} x)$. Then

$$\begin{aligned} \tilde{h}_l(x) &= \sigma(\tilde{\mathbf{w}}^{(l)} \tilde{h}_{l-1}(x)) = \sigma(P_l \mathbf{w}^{(l)} P_{l-1}^{-1} P_{l-1} h_{l-1}(P_0^{-1} x)) \\ &= \sigma(P_l \mathbf{w}^{(l)} h_{l-1}(P_0^{-1} x)) = P_l \sigma(\mathbf{w}^{(l)} h_{l-1}(P_0^{-1} x)) \\ &= P_l h_l(P_0^{-1} x). \end{aligned}$$

i.e. $\tilde{h}_l(x) = P_l h_l(P_0^{-1} x)$. So for the output layer $\tilde{\mathbf{w}}^{(L+1)} = P_{L+1} \mathbf{w}^{(L+1)} P_L^{-1} = \mathbf{w}^{(L+1)} P_L^{-1}$, one has

$$\begin{aligned} f(x, g \cdot \mathbf{w}) &= f(x, \tilde{\mathbf{w}}) = \tilde{\mathbf{w}}^{(L+1)} \tilde{h}_L(P_0^{-1} x) \\ &= \mathbf{w}^{(L+1)} P_L^{-1} P_L h_L(P_0^{-1} x) \\ &= \mathbf{w}^{(L+1)} h_L(P_0^{-1} x) \\ &= f(P_0^{-1} x, \mathbf{w}). \end{aligned}$$

This clearly does not naturally lead to G' -invariance of the supervised-learning loss in Equation (1). However, we can instead consider the teacher-student loss in (13). Now let $\boldsymbol{\eta} = P_0^{-1}x$, since $x \sim q(x)$ is S_{k_0} invariant, i.e. $h \cdot x \sim q(x)$, $h \in S_{k_0}$, then $\boldsymbol{\eta} = P_0^{-1}x$ also follows $q(x)$ distribution. Moreover, since teacher model f_T has input permutation invariance i.e. $f_T(h \cdot x, \mathbf{w}_T) = f_T(x, \mathbf{w}_T)$, for $h \in S_{k_0}$. Therefore, the loss function takes the form

$$\begin{aligned} L_{ts}(g \cdot \mathbf{w}) &:= \mathbb{E}_{P_0 \boldsymbol{\eta} \sim q(x)} \|f(\boldsymbol{\eta}, \mathbf{w}) - f_T(P_0 \boldsymbol{\eta}, \mathbf{w}_T)\|^2 \\ &= \mathbb{E}_{\boldsymbol{\eta} \sim q(x)} \|f(\boldsymbol{\eta}, \mathbf{w}) - f_T(\boldsymbol{\eta}, \mathbf{w}_T)\|^2 = L_{ts}(\mathbf{w}). \end{aligned}$$

□

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